

INCH-POUND YACL-WP PD 99-01 20 April 2001
--

PURCHASE DESCRIPTION

COVERALLS, FLYER'S, MEN'S, SUMMER, FLAME-RESISTANT

1. Scope. This specification covers requirements for the CWU-27/P coverall to be worn as a protective garment by flight personnel.

1.2 Classification. The coverall shall be furnished in the following types, lengths and sizes, as specified (see 6.2):

Class 1- Sage Green 1590

Class 2- Tan 380

1.3 Sizes: The coveralls shall be furnished in even sizes 32 to 52, in Short, Regular and Long.

APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4. This section does not include documents cited in other sections or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements and documents cited in sections 3 and 4, whether or not they are listed.

2.1 Government Documents

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-55634	Fastener, Slide, Interlocking
A-A-55217	Thread, Aramid, Spun Staple

MILITARY

MIL-F-10884	Fasteners, Snap
MIL-DTL-83429	Cloth, Plain and Basket Weave, Aramid

(Unless otherwise indicated, copies of Federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government Publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the specification. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

ANSI/ASQC Z1.4 Sampling Procedures and Tables for Inspection of Attributes.

Application for copies should be addressed to American National Standards Institute, 11 West 42nd Street, New York, NY 10036-8002.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) PUBLICATION

ASTM-D-3776 Standard Test Method for Mass per Unit Area Weight of Fabric
ASTM-D-5169 Standard Test Method for Shear Strength (Dynamic method) of Hook and Loop Touch Fasteners
ASTM-D-5035 Breaking Force and Elongation of Textile Fabrics (Strip Method)
ASTM-D-6193 Standard Practice for Seams and Stitches
ASTM-D-5170 Standard Test Method for Peel Strength ("T" method for Hook & Loop Touch Fasteners)

(Applications for copies should be addressed to American Society for Testing Materials, 100 Barr Harbor Drive, Conshohocken, PA 19428)

AMERICAN SOCIETY OF TEXTILE CHEMISTS AND COLORISTS (AATCC) PUBLICATION

AATCC-D-88 Smoothness of Seams after Home Laundering
AATCC-61 Colorfastness to Laundering, Home and Commercial: Accelerated

(Applications for copies should be addressed to American Association of Textile Chemists and Colorists (AATCC) P.O. Box 12215, Triangle Park, NC 27709-2215)

AEROSPACE INDUSTRY ASSOCIATION OF AMERICA INC.

NASM16491 Grommet, Metallic

(Applications for copies should be addressed to Aerospace Industry Association of America Inc., 1250 Eye Street, N.W. Suite 1100, Washington, DC 20005.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Description

The coveralls shall be a one-piece garment with a slide fastener front closure, a bi-swing back, hanger loop, shoulder patches, loop fastener name plate, hook and loop fastener tape adjustment tabs for the waist and sleeve, and a slide fastener closure on each lower leg. The front closure has beaded edges backed with a protective fly. The waistband shall have elastic webbing in the center back. The coveralls shall have two slanted breast pockets, one combination utility and two pencil compartment pocket with a pencil pocket flap on the upper left sleeve, two thigh patch pockets, a reinforced knife pocket with lanyard on the left thigh, and two lower leg patch pockets. Except for the knife pocket, all pockets shall have slide fastener covers with beaded edges. (Reference figures 1. and 2.)

3.3 Material.

3.3.1 Basic Material. The basic cloth shall be plain weave, flame resistant, and shall meet the requirements of Type II, class 6 in MIL-C-83429B. The color of the finished cloth shall be Sage Green 1590 (Class 1) or Tan 380 (Class 2) and shall match the standard sample.

3.3.2 Thread. The thread for all sewing operations shall conform to Type I of A-A-55217 and shall have a minimum elongation of 20% which is comparable to Nomex.TM. The color shall be a good match to the basic material. The nominal Tex size of the needle thread shall be 35 for all operations. The color shall be a good match to the basic material. The thread tension shall be properly maintained so that stitching will not be too tight or loose.

3.3.3 Hook and Loop Fastener Tape. The hook and loop fastener tape color shall be an approximate match to the basic material with a 6.5 mil nylon hook (220 denier) and nylon ground and shall meet the requirements in Table I. The hook and loop tapes shall be stabilized as necessary to allow maximum flatness and dimensional stability. Hook and loop shall show no signs of fraying after cutting and attachment to coveralls. The sheer strength of the hook shall be 5.0 grams/linear yard, the peel strength of the hook shall be .5 grams/linear yard, and the weight shall be 4.5 grams/linear yard for the hook and 5.9 grams/linear yard for the loop when tested in accordance with paragraph 4.3.2.

Table I.					
Hook and Loop Fastener Tape Lengths (inches)					
Pencil Pocket Hook & Loop	Waist Hook <u>1</u> /	Waist Loop <u>1</u> /	Sleeve Hook <u>2</u> /	Sleeve Loop <u>2</u> /	Nameplate Hook & Loop
<i>all sizes</i>	<i>all sizes</i>	<i>sizes:</i>	<i>all sizes</i>	<i>all sizes</i>	<i>all sizes</i>
1 by 2	1 by 4	30 - 34 1 by 5 36 - 38 1 by 6 40 - 44 1 by 7	1-1/2 by 2	1-1/2 by 6	2 by 4

1/ Each 2/ Each sleeve

3.3.4 Elastic Webbing. The elastic webbing for the waistband section shall be 1-1/2 inches wide, with a minimum thickness of 0.035 to 0.040 inches. The cut length shall be as referenced in Table II. After the cut length of the webbing has been stretched and maintained at 50% elongation, and then allowed to relax for 10 minutes, the webbing tension set (change in length of the sample) shall not exceed 5%.

Table II	
Elastic Lengths (inches)	
All sizes, all lengths	7

3.3.5 Metallic Grommet. The metallic grommet for the knife pocket shall conform to Type 1, class 3, Size No.0 of NASM16491.

3.3.6 Nylon Cord. The natural color nylon braided cord for the lanyard shall be 60 inches long with the ends seared. The cord shall be rolled into a coil 4 inches long and enclosed in the knife pocket. The diameter of the cord shall be 3/32 inch. The end of the cord shall be dipped to prevent fraying. The cord shall have a minimum breaking strength of 95 pounds when tested as specified in 4.3.1.

3.3.7 Reinforcement Webbing. The synthetic woven webbing used for the reinforcement piece of the knife pocket shall be 1-1/2 inches wide, .85 ounces per linear yard (max.) .025 to .040 inch. thick, with a breaking strength of 1200 lbs. min. when tested as specified in 4.3.1.

3.3.8 Snap Fastener. The snap fastener for the knife pocket shall conform to style 2, finish 2 of MIL-F-10884 and MS27980-1B (button), MS27980-6B (socket), MS27980-7B (stud) and MS27980-8B (eyelet).

3.3.9 Binding Tape. The strip for binding the edges of the knife pocket shall be bias-cut, 1-1/8 inch wide, from the basic fabric. The finished width shall be no more than 3/8 inch wide.

3.3.10 Slide Fasteners. The slide fasteners shall conform to the requirements of A-A-55634 and to tables III and IV, of this document. The chain and other components except the latch and

spring of the automatic lock shall be brass with a black chemical finish. The front opening slide fastener shall have a thong made of double thickness of the basic material. The tape of the slide fasteners shall be 3/4 inch wide, high temperature resistant polyamide, and shall approximately match the color of the basic material. The hole in the pull tab shall be not larger than 1/4 inch in width and 3/8 inch in length. The slide fastener shall have a 1 to 1-1/4 inch top tape extension and a 1-1/2 to 1-3/4 bottom tape extension. All other slide fasteners shall have a short tab pull, and a 1 to 1-1/4 inch top and bottom tape extension. Lengths of slide fasteners in Table III shall be measured from stop to stop.

Table III Slide Fasteners						
Location	Quantity	Type	Style	Size	Sliders	Direction slide fasteners close
Front Opening	1	I	13	7-8	2	Top slider upward and bottom slider downward
Leg Openings	2	I	4	5-7	1	Downward
Breast Pockets	2	I	7	5-7	1	Downward (toward center)
Lower Leg Pockets	2	I	7	5-7	1	Closes toward the back of the leg (figure 2)
Left Thigh Pocket	1	I	7	5-7	1	Downward
Right Thigh Pocket	1	I	7	5-7	1	Direction of inseam
Left Sleeve Pocket	1	I	7	5-7	1	Upward

Table IV Length of slide fasteners (in inches).

Coverall size	Front opening	Leg openings	Breast pockets	Thigh pockets	Lower leg Pockets	Left sleeve pockets
32S	25	10	6 1/2	7	8-1/2	4-3/4
32R	26	10	6 1/2	7	8-1/2	4-3/4
32L	27	10	6 1/2	7	8-1/2	4-3/4
32XL	28	10	6 1/2	7	8-1/2	4-3/4
32XXL	29	10	6 1/2	7	8-1/2	4-3/4
34S	25-1/2	10	6 1/2	7	8-1/2	4-3/4
34R	26-1/2	10	6 1/2	7	8-1/2	4-3/4
34L	27-1/2	10	6 1/2	7	8-1/2	4-3/4
34XL	28-1/2	10	6 1/2	7	8-1/2	4-3/4
34XXL	29-1/2	10	6 1/2	7	8-1/2	4-3/4
36S	26	10	6 1/2	7	8-1/2	4-3/4
36R	27	10	6 1/2	7	8-1/2	4-3/4
36L	28	10	6 1/2	7	8-1/2	4-3/4
36XL	29	10	6 1/2	7	8-1/2	4-3/4
36XXL	30	10	6 1/2	7	8-1/2	4-3/4
38S	26-1/2	10	6 1/2	7	8-1/2	4-3/4
38R	27-1/2	10	6 1/2	7	8-1/2	4-3/4
38L	28-1/2	10	6 1/2	7	8-1/2	4-3/4
38XL	29-1/2	10	6 1/2	7	8-1/2	4-3/4
38XXL	30-1/2	10	6 1/2	7	8-1/2	4-3/4
40S	27	10	6 1/2	7	8-1/2	4-3/4
40R	28	10	6 1/2	7	8-1/2	4-3/4
40L	29	10	6 1/2	7	8-1/2	4-3/4
40XL	30	10	6 1/2	7	8-1/2	4-3/4

Table IV Length of slide fasteners (in inches).

Coverall size	Front opening	Leg openings	Breast pockets	Thigh pockets	Lower leg Pockets	Left sleeve pockets
40XXL	31	10	6 1/2	7	8-1/2	4-3/4
42S	27-1/2	10	6 1/2	7	8-1/2	4-3/4
42R	28-1/2	10	6 1/2	7	8-1/2	4-3/4
42L	29-1/2	10	6 1/2	7	8-1/2	4-3/4
42XL	30-1/2	10	6 1/2	7	8-1/2	4-3/4
42XXL	31-1/2	10	6 1/2	7	8-1/2	4-3/4
44S	28	10	6 1/2	7	8-1/2	4-3/4
44R	29	10	6 1/2	7	8-1/2	4-3/4
44L	30	10	6 1/2	7	8-1/2	4-3/4
44XL	31	10	6 1/2	7	8-1/2	4-3/4
44XXL	32	10	6 1/2	7	8-1/2	4-3/4
46S	28-1/2	10	6 1/2	7	8-1/2	4-3/4
46R	29-1/2	10	6 1/2	7	8-1/2	4-3/4
46L	30-1/2	10	6 1/2	7	8-1/2	4-3/4
46XL	31-1/2	10	6 1/2	7	8-1/2	4-3/4
46XXL	32-1/2	10	6 1/2	7	8-1/2	4-3/4
48S	29	10	6 1/2	7	8-1/2	4-3/4
48R	30	10	6 1/2	7	8-1/2	4-3/4
48L	31	10	6 1/2	7	8-1/2	4-3/4
48XL	32	10	6 1/2	7	8-1/2	4-3/4
48XXL	33	10	6 1/2	7	8-1/2	4-3/4
50S	29-1/2	10	6 1/2	7	8-1/2	4-3/4
50R	30-1/2	10	6 1/2	7	8-1/2	4-3/4
50L	31-1/2	10	6 1/2	7	8-1/2	4-3/4
50XL	32-1/2	10	6 1/2	7	8-1/2	4-3/4
50XXL	33-1/2	10	6 1/2	7	8-1/2	4-3/4
52S	30	10	6 1/2	7	8-1/2	4-3/4
52R	31	10	6 1/2	7	8-1/2	4-3/4
52L	32	10	6 1/2	7	8-1/2	4-3/4
52XL	33	10	6 1/2	7	8-1/2	4-3/4
52XXL	34	10	6 1/2	7	8-1/2	4-3/4
Tolerance	±3/8	±3/16	±3/16	±3/16	±3/16	±3/16

3.4 Labels. Each coverall shall have a combination identification, size, and instruction label. The label shall be stitched flat on all four sides in the inside center back of the garment approximately 1 inch below the hanger. The label shall contain the item name (including CWU-27/P, Type I or Type II, Class 1 or Class 2, document number, contractor's name, contract number and national stock number (NSN) and fiber content. The inscription shall have a minimum font size of 10. The inscription legibility, label, and label attachment shall last the expected life of the coveralls. The size shall be marked with a numerical size (38, 40 etc.), and S, R, L for short, regular or long. The instructions shall contain the following information:

1. Launder with slide fasteners and hook and loop engaged. Hand or machine wash in warm water using a mild detergent. Permanent press cycle. Flat dry or

machine dry on low heat or permanent press cycle. REMOVE PROMPTLY. DO NOT PRESS HOOK AND LOOP FASTENERS. DO NOT DRY-CLEAN.

2. USE DETERGENTS ONLY. DO NOT USE STARCH, SOAP, BLEACH, FABRIC SOFTENER OR DRYER SHEETS. THESE PRODUCTS WILL DETERIORATE THE FLAME RESISTANCE.
3. Lubricate slide fasteners with wax pencil.

DO NOT REMOVE THIS LABEL

3.4.1 Barcode tag. Each item shall be individually bar-coded, clearly legible and readable by scanner. The bar-coding element shall be a 13 digit national stock number (NSN). The bar code type shall be medium to high code density, and shall be attached so that it is completely visible on the item when it is folded and /or packaged as specified. The label shall be attached without using adhesive or without piercing the fabric to prevent damage to the item.

3.5 Patterns. The government shall furnish a complete set of patterns or a master pattern with grade rules, to maintain uniformity and consistency in manufacturing. The government patterns shall be used to create the contractor's working patterns. Minor pattern modifications are permitted to accommodate manufacturing procedures, however the design and finished measurements shall be maintained. Pattern grain lines shall not be modified. Adhesive patterns are prohibited.

Table V. List of pattern parts

<u>Pattern piece</u>	<u>Nomenclature</u>	<u>Cut parts</u>
1. Right front	RFRONT	1
2. Left Front	LFRONT	1
3. Front Facing	FRONTFACING	2
4. Top back	UPPERBACK	1
5. Lower back	LOWERBACK	2
6. Upper Collar	TOPCOLLAR	1
7. Under Collar	UNDERCOLLAR	1
8. Collar interlining	CLRINTRLINE	1
9. Shoulder Patch	SHOULDER	2
10. Back belt casing	BELTBACK	1
11. Right Top Sleeve	RTOPSLEEVE	1
12. Left Top Sleeve	LTOPSLEEVE	1
13. Under Sleeve	UNDERSLEEVE	2
14. Breast Pocket	BREASTPOCKE	2
15. Breast Pocket Slide Fastener cover	BREASTCOVER	2
16. Sleeve pocket	SLEEVEPKT	1
17. Sleeve pckt Slide fastener Cover	SLVPKTCOVER	1
18. Pencil pocket	PENCILPOCKE	1
19. Pencil Pocket Flap	PENCILFLAP	1
20. Right Thigh Pocket	RTHIGHPOCKE	1
21. Right Thigh Pckt Slide fastener cover	RTHIGHCOVER	1
22. Left Thigh Pocket 32-36	LTHIGHPKT_S	1

23. Left Thigh Pocket	38-44	LTHIGHPKT_M	1
24. Left Thigh Pocket	46-52	LTHIGHPKT_L	1
25. Left Thigh Pckt Slide Fastener Cover		LTHIGHCOVER	1
26. Lower Leg Pocket		LEGPOCKET	2
27. Lower Leg Pckt Slide Fstnr Cover		LEGCOVER	2
28. Knife Pocket		KNIFEPOCKET	1
29. Knife Backing			
a. 32-40		KNIFEBACK_S	1
b. 42-52		KNIFEBACK_L	
30. Knife Flap		KNIFEFLAP	1
31. Leg Slide Fastener Cover		GUSSETCOVER	4
32. Front Slide Fastener cover		ZIPPERCOVER	1
33. Protective Fly		FLY	2
34. Front Belt Tab		WAISTADJUST	2
35. Sleeve Tabs		SLVADJ	2
36. Thong		THONG	1
37. Hanger loop		HANGER	1

Pattern guides

38. Breast pocket guides	BRSTPKTGUID	1
39. Leg Pocket Guides	LEGPKTGUIDE	1

3.6 Figures. The figures are furnished for information purposes only. When inconsistencies exist between the written specification and the figures, the written purchase description shall govern.

3.7 Manufacturing Operations Requirements. The coveralls shall be manufactured in accordance with operation requirements specified in Table VII. The contractor is not required to follow the exact sequence of operations provided the finished coverall is identical to that produced by following the sequence as listed in Table VII. Minor modifications are permitted where necessary when using automated equipment. These modifications shall not alter the dimensional, serviceability or appearance requirements cited in the specification.

3.7.1 Stitching. The thread tension shall be properly maintained so that stitching will not be too tight or loose.

3.7.1 Workmanship. The coveralls shall be free from all defects which would affect proper functioning in service.

TABLE VII Table of Operations

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
1.	<u>Cutting</u> a. Cut the coverall in strict accordance with the patterns. Place the patterns on the material with the directional lines coinciding with the warp direction. The directional line may not vary. b. Cut all parts from one piece of material except the sleeve tabs, waist tabs, front facings, undercollars, hanger, protective fly, and the front slide fastener cover, which may be cut from ends. However, all pieces must be cut in accordance with directional lines on the patterns. c. Replace any parts containing holes or seriously weakening defects. Metal fastening devices or sewn on tickets shall not be used for marking.			
2.	<u>Overedging</u> a. Overedge all slide fastener covers and pocket raw edges that are not on the selvage.	502, 503 or 504	Ssa-1	8-10
3.	<u>Collar</u> a. Sew the undercollar and collar interlining together with diagonal stitching with the points of the stitching approximately 3 inches apart as indicated by patterns. b. Join the collar to the undercollar around the outer edge with a single row of stitching 1/4 inch from the edge. c. Turn collar, forcing out edges.	301 or 401 301	SSv-1 Sse-2 (a)	10-12 10-12
4.	<u>Sleeve and Waist Tabs</u> a. Fold tabs lengthwise with right sides together and stitch 1/4 inch from the edge leaving one square end open. b. Turn, work out edges, and topstitch 1/4 inch from the edge. c. Position the hook fastener tape on the tabs as indicated by the pattern marks and stitch on all four sides 1/16 to 1/8 inch from the edge.	301 301 301	SSe-2 (a) SSe-2 (b) SSa-1	10-12 10-12 10-12

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
	Note: Two piece tab construction is an acceptable alternative.			
5.	<u>Corded beading in edges of slide fastener covers</u> NOTE: All slide fastener closures shall have corded beading in the folded edge of the covering. The beaded edges should be flush with each other when the slide fastener is closed. a. For all slide fastener closure edges, including the center front opening, bead the edge by placing the cord at the fold line, folding cloth wrong sides together, and stitching close to the cord.	301	SSbg-3a	10-12
6.	<u>Sleeve Pocket</u> Note: The pocket shall be a bellowed compartment with a covered vertical slide fastener closure (on the forward side of the full length of the pocket) with two pencil compartment openings top stitched to the sleeve compartment and covered with a pencil pocket flap which fastens with hook and loop tape. The openings shall be 7/8 inch wide by 6 3/8 inches long $\pm$ 1/8 inch. The pocket shall finish 5-3/4 inches long by 4-1/4 inches wide. (The width measurement is taken across the top of the pocket, including the slide fastener cover.) a. Hem the top of the pencil pocket compartment by folding the 3/8 inch seam allowance to the wrong side and top stitching with the first row 1/16 inch from the edge. b. Position the loop fastener tape on the pocket as indicated by the pattern marks and stitch on all four edges 1/16 to 1/8 inch from the edge. c. Double topstitch the pencil pocket to the sleeve pocket; matching top corners to drill markings and stitching down each side. The edges of the pocket should be turned under 3/8 inch and the bottom raw edge of the pencil pocket will line up with the bottom raw edge of the sleeve pocket. The top of the pocket will have ease to form pencil sleeves (see next operation) and the bottom edge will be flat to the sleeve pocket. d. Form the individual pencil pockets by stitching down the center of the pencil pocket with a single row of stitching. The pockets when stitched should form a semicircular 7/8 inch opening that a pen/pencil slides into easily. e. Bartack the ends of the pencil pocket openings with the bartack superimposed on the inner row of double stitching and on the center row of stitching. The bartack shall be 3/8 to 1/2 inch	301 301 301 301 bartack	EFa-2 SSa-1 LSd-2 SSv-1	10-12 10-12 10-12 10-12 bartack

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
	long. Note: the bartacks superimposed on the stitch lines must not be placed totally to the inside of the stitch line. This will reduce the dimension of the pencil pockets rendering them nonfunctional. f. Stitch the bellow seams at the bottom of the sleeve pocket. g. Sew the slide fastener tape to the forward side of the sleeve pocket with a double row of stitching $\frac{1}{4}$ inch from the edge of the beading. The beading should cover the slide fastener teeth. h. Sew the slide fastener cover to the opposite side of the remaining slide fastener tape using a single row of stitching $\frac{1}{4}$ inch from the bead stitching. The beaded edge should cover the slide fastener teeth. i. On the pencil pocket flap, position the hook fastener tape on the flap as indicated by the drill holes and stitch on all four edges $\frac{1}{16}$ to $\frac{1}{8}$ inch from the edge. j. Fold pencil pocket flap, right sides together, and seam the two long side edges with a single row of stitching. k. Turn pencil pocket flap, work out corners, and topstitch all finished edges $\frac{1}{16}$ to $\frac{1}{8}$ inch from the edge. l. Stitch the sleeve pocket assembly to the left sleeve as indicated by pattern marks $\frac{1}{16}$ inch from the edge, incorporating the unfinished edge of the pencil pocket flap. The edge of the pocket should be turned under $\frac{3}{8}$ inch. The pencil pocket flap should line up with the pencil pocket and be placed for stitching with the fastener tape up. NOTE: The six drill marks indicate the four corners of the assembled pocket and the top and bottom line where the pocket cover meets the pocket. The assembled pocket slightly covers the drill marks. m. Place a second row of stitching $\frac{1}{4}$ inch from the front edge and top edge of the pocket. Fasten the flap to the pocket. n. Bartack across each end of the slide fastener on the inner row of topstitching on the top, and $\frac{1}{4}$ inch below the bottom stop.	301 301 301 301 301 301 301 bartack	SSa-1 LSb-2 SSa-1 SSa-1 SSa-1 LSd-2 LSd-2	10-12 10-12 10-12 10-12 10-12 10-12 10-12 bartack
7.	<u>Right Thigh pocket</u>			

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
	a. Sew the slide fastener to the top of the pocket with a single row of stitching 1/4 inch from the bead stitching. The beading should cover the slide fastener teeth. b. Sew the slide fastener cover on the opposite side of the tape with a single row of stitching 1/4 inch from the edge of the bead stitching. The beaded edge shall cover the teeth. c. Matching the notches on the side seam with the top and bottom of the assembled pocket double topstitch the pocket to the right front at the top, bottom and front edge. The side edge will be stitched with the side seam. The edges of the pocket shall be turned under 3/8 inch.	301	LSb-2	10-12
		301	SSa-1	10-12
		301	LSd-2	10-12
8.	<u>Left Thigh Pocket</u> a. Sew the slide fastener to the forward side of the pocket with a single row of stitching 1/4 inch from the bead stitching. The beaded edges shall cover the scoops. b. Sew the slide fastener cover to the opposite side of the tape with a single row of stitching 1/4 inch from the edge of the bead stitching. The beaded edge shall cover the teeth. c. Matching the notches on the side seam with the top and bottom of the assembled pocket, double topstitch the pocket to the left front at the top, bottom and front edge. The side will be stitched with the side seam. The edges of the pocket should be turned under 3/8 inch.	301	LSb-1	10-12
		301	SSa-1	10-12
		301	LSd-2	10-12
9.	<u>Lower Leg Pocket Preparation</u> a. Sew the slide fastener to the top of the pocket with a single row of stitching 1/4 inch from the edge of the bead stitching with the beaded edges covering the teeth. b. Sew the slide fastener cover on the opposite side of the tape stitching 1/4 inch from the edge of the bead stitching. The beaded edge should cover the slide fastener teeth.	301	LSb-1	10-12
		301	SSa-1	10-12
10.	<u>Knife Pocket</u>			

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
	a. Place the webbing reinforcement on the pocket material wrong side up, as indicated by the drill holes, stitch 1/16 inch from the edge on all four sides.	301	SSa-1	10-12
	b. Fold the pocket lengthwise and bind the square end with a 1-1/8 wide, bias cut shell material binding.	301	BSc-1	10-12
	c. On the small flap reinforcement piece for the top of the pocket, bind the straight edge with a 1-1/8 inch wide, bias cut, shell material binding. The binding shall finish no more than 3/8 inch wide.	301	BSc-1	10-12
	d. Position the flap reinforcement piece at the round end of the pocket on the same side as the webbing piece. Hold in place through both pocket layers with a single row of stitching 1/8 inch from the edge.	301	SSa-1	10-12
	e. Fold the pocket material at the fold line with the reinforcement webbing on the inside. Leaving a 1/2 inch tail beyond the pocket end, bind the raw edges with a 1-1/8 inch wide, bias cut shell material binding starting and stopping at the bottom of the pocket.	301	BSc-1	10-12
	f. Form a pleat on the flap reinforcement piece by folding its excess flat and stitching across its width approximately 1 inch from the straight bound edge.	301	LSa-1	10-12
	g. Install a size O grommet on the pocket, at the drill hole located below the bottom edge of the flap reinforcement piece.			
	h. With edges turned under 3/8 inch, double top stitch the knife backing patch to the left front leg where indicated by notches.	301	LSd-2	10-12
	i. Double topstitch the knife pocket to the knife backing patch on the left leg on the long side edges and the bottom on the binding. Position in accordance with the drill holes, with the opening on the inside. Binding tails should be turned under on each side and caught in the stitching. Backstitch the ends of the stitching 1/2 inch on each side of the opening.	301	LSd-2	10-12
	j. On the bound square edge of the knife pocket opening, secure the top of the pocket to the patch/leg with a single row of stitching.	301	SSa-1	10-12
	k. Stamp one snap fastener socket to the center of the rounded pocket flap as indicated by the drill hole.			

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
	l. Stamp one snap fastener stud, reinforced on the inside with a piece of the shell cloth, on the coveralls to correspond with the snap fastener socket. m. Pull a 60 inch long nylon cord through the grommet. Tie the resulting lanyard with a square knot. Sear both ends of the nylon cord (or the ends may be dipped in a suitable resin). Fold the loose cord in layers approximately 4 inches long and position the folded cord in the pocket, parallel with the pocket, with all the folds inside the pocket. Snap the flap closed.			
11.	<u>Front Protective Fly</u> a. With right sides together, stitch both plies of the protective fly along the outer edge with a 3/8 inch seam allowance. b. Turn the fly. Work out the edges and topstitch with a single row of stitching 1/4 inch from the seamed edge. c. Double topstitch the fly down the center of its entire length.	301 301 301	SSe-2 (a) SSe-2 (b) SSv-2	10-12 10-12 10-12
12.	<u>Thong for front slide fastener.</u> a. Fold the thong lengthwise in accordance with the notches to finish 1/2 inch wide. Stitch to secure. Folders may be used to fabricate strips.	301	EFp-2	10-12
13.	<u>Center front slide fastener</u> <i>NOTE: The slide fastener shall start 1/2 ($\pm$ 1/4 inch) inch from the neckline seam and the protective fly must completely cover the slide fastener.</i> a. Place the protective fly on the right side of the front facing with the finished edge of the protective fly 3/8 inch down ($\pm$ 1/8 inch) from the cut edge of the neck of the front facing with the raw edges even. Join with a single row of stitching 3/8 inch from the raw edge. b. Turn under one short end of the slide fastener cover piece 1/4 inch. With a single row of stitching 1/16 inch from the fold, stitch the cover to the bottom of the backside of the slide fastener just below the stop. c. Position the right slide fastener tape on the protective fly/front facing so that the raw edge of the center front seam allowance is	301 301 301	SSa-1 LSd-1 LSq-2a	10-12 10-12 10-12

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
	even with the edge of the tape. Join with a single row of stitching, 1/4 inch from the slide fastener teeth. The beaded edge of the center front should just cover the slide fastener teeth. d. Fold the facing over the stitching in 13.c and topstitch 1/16 inch from the folded edge, catching the coveralls, slide fastener tape, protective fly and the facing. When executed correctly, on the right side of the coveralls this stitching should appear to be topstitched approximately 1/4 inch from the bead stitching at the center front. e. Turn the raw edge of the facing under 1/2 inch and topstitch to the coverall front 1/16 inch from the folded edge. f. Repeat steps 13.c through 13.e for the left slide fastener tape, except that there is no protective fly. g. Pull the thong through the hole in the top of the slide fastener pull. With the raw edges even, bartack the bottom, center and up close to the bottom pull across the 1/2 inch width.	301 301 bartack	LSq-2b LSd-1	10-12 10-12 bartack
14.	<u>Right and left breast pockets</u> a. Sew the slide fastener tape to the top diagonal side of the pocket with a single row of stitching, 1/4 inch from the bead stitching. The beading shall cover the slide fastener teeth. b. Sew the slide fastener cover to the opposite slide fastener tape using a single row of stitching 1/4 inch from the bead stitching. The beaded edge shall cover the slide fastener teeth. c. For pocket placement match the following: (1) Notches in sides of pockets with notches in side seams, (2) Bottom side corners of pockets with back waist seam, (3) drill markings on fronts to inside corners of assembled pockets; then, with the top, bottom and front edges turned under 3/8" inch, double topstitch these edges to the fronts. The side edges will be stitched with the side seams. Note: A breast pocket guide (template) is provided as an alternate method for pocket placement.	301 301 301	LSb-1 SSa-1 LSd-2	10-12 10-12 10-12
15.	<u>Wearers nameplate</u> Position the loop part of the nameplate parallel to the center front beaded edge and offset 1 inch so that the tape is above, but touching, the left breast pocket. Stitch on all four side 1/16 to 1/8	301	LSbj-1	10-12

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
	inch from the edge.			
16.	<u>Upper back: Bi-swing pleats</u> a. Fold the upper back with right sides together from the middle pleat notch on the shoulder to the middle pleat notch at the waist. Stitch 1/16 on the fold from shoulder to waist. Repeat for the other side. b. Fold the upper back with wrong sides together from the notch nearest the center back on the shoulder to the notch nearest the center back at the waist. Double topstitch on the fold from shoulder to waist. Repeat for other side.	301	OSf-1	10-12
17.	<u>Seat seam</u> a. Join the lower back seam with a double-lapped, double stitched seam. The right panel shall overlap the left panel.	301 or 401	LSc-2	10-12
18.	<u>Joining of upper back to lower back.</u> Join with a double-lapped, double-stitched seam. The upper back shall overlap the lower back.	301 or 401	LSc-2	10-12
19.	<u>Shoulder seams</u> Join with a double-lapped, double-stitched seam. The back shall overlap the front.	301 or 401	LSc-2	10-12
20.	<u>Shoulder patch</u> a. Pattern is annotated to mark front. Matching the notches and their connecting lines to the shoulder seam, double topstitch the shoulder patch to the coverall on all edges except the armhole edge which will be stitched with the armhole seam. The edges of the patch should be turned under 3/8 inch.	301	LSd-2	10-12
21.	<u>Sleeves</u> a. For underarm ventilation make sewn eyelets on sleeves as indicated by pattern marks. Finished inside diameter should be 1/8 to 3/16 inch.	401		19to24 stitches per

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
	b. Double-lap, double-stitch the elbow seams. The topsleeve should overlap the undersleeve. c. Double-lap, double-stitch the armhole seams. The front and back panels should overlap the sleeves. Note: the notch in upper back matches sleeve seam. d. Hem the sleeves. Turn under 3/8 inch and then 1-1/4 inch from the hem. Secure with a single topstitch 1/16 inch from the edge. e. Position the loop fastener tape on the top sleeve as indicated by the pattern marks and stitch on all four sides 1/16 to 1/8 inch from the edge.	301 or 401 301 or 401 301 301	LSc-2 LSc-2 EFb-1 SSa-1	eyelet 10-12 10-12 10-12 10-12
22.	<u>Sleeve tabs</u> Position each sleeve tab between the notches on the top sleeve with the raw edges even. Hold in place by stitching 1/8 inch from the cut edge.	301	LSd-2	10-12
23.	<u>Side seams</u> Double-lap, double-stitch the side seams from the wrist to the ankle. The backs shall overlap the fronts.	301	LSc-2	10-12
24.	<u>Joining of lower leg pocket</u> Double topstitch lower leg pockets to lower legs in accordance with the leg pocket guide. The edge of the pockets should be turned under 3/8 inch. The pocket will finish 5-5/8 inches up from the cut edge of the hem at the side seam. (4 inches up from the finished hem)	301	LSd-2	10-12
25.	<u>Leg Hems</u> Turn raw edge under 3/8 inch and then 1-1/4 inch for the hem. Topstitch 1/16 inch from the edge.	301	EFb-1	10-12
26.	<u>Gusset</u> Fold the leg bottom along the centerline matching the pattern marks and topstitch 1/16 inch from the folded edge to create the gusset. Note: The gusset can be sewn before the hem (operation 25) if desired.	301	OSf-1	10-12
27.	<u>Leg Slide Fasteners</u>			

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
	a. Double stitch the slide fastener tapes to the notched side of the lower leg slide fastener cover. The beading shall cover the slide fastener teeth. The first row shall be 1/4 inch from the bead stitching. b. Repeat 26(a) on the opposite slide fastener tape with another cover. c. Repeat 26(a) and 26(b) for the other leg slide fastener covers. d. Double topstitch the leg slide fastener to the lower legs in accordance with marks on pattern. The edges of the covers should be turned under 3/8 inch. The finished edge of the covers should end at the hemline.	301	LSd-2	10-12
		301	LSd-2	10-12
28.	<u>Belt Assembly</u> a. To secure the waist adjustment tab to the elastic, fold the edge of the waist adjustment tab under 3/8 inch and position the fold 1/2 inch from an end of the elastic. Stitch a box X, 3/8 inch by 7/8. Repeat with other tab on other end of the elastic. b. Center the elastic on the center back of the waist seam. Stitch a vertical bartack through the elastic and the coveralls at the intersection of the waist and seat seams. c. Double fold the short ends of the back belt to the inside 1/4 inch and topstitch 1/16 inch from the last fold. d. Double topstitch the two long edges of the back belt to the coveralls back. The edges of the belt should be turned under 3/8 inch. The belt should be centered over the waist seam. e. Position the loop waist fastener tape to the right and left fronts centered over the lower edge of the breast pockets, extending from the side seam toward the front edge of breast pocket, and stitch on all four sides 1/16 to 1/8 inch from the edge. The fastener tape should be aligned with the waist adjustment tabs.	301	LSd-1	10-12
		bartack		28 per bartack
		301	EFb-1	10-12
		301	LSd-2	10-12
		301	LSbj-1	10-12
29.	<u>Inseam</u> Double-lap double-stitch the inseams with the backs overlapping the fronts.	301 or 401	LSc-2	10-12
31.	<u>Joining of collar to neck</u>			

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
	a. Sew the undercollar to the neck with a single row of stitching 3/8 inch from the edge.	301	SSa-1	10-12
	b. Turn under the raw edge and top stitch the top collar to the neckline seam with a single row of stitching 1/16 inch from the folded edge.	301	LSd-1	10-12
	c. Topstitch the collar 1/4 inch from the edge with a single row of stitching.	301	SSe-2 (b)	10-12
32.	<u>Hanger loop</u> a. Fold the hanger loop lengthwise in accordance with the notches to finish 1/2 inch wide by 4 inches long. Stitch to secure. b. Position the hanger loop on the inside center of the back to the neck, $1 \pm 1/4$ inch below the neckline seam, and secure at the ends with double stitching or bartacks.	301	Efp-2	10-12
		301	LSd-2	10-12
33.	<u>Labels</u> Position the labels approximately 1 inches from the neck seam, centered directly below the hanger loop, on the inside back of the coverall. Stitch on all four sides 1/16 inch from the edge.	301	SSa-1	10-12
34.	<u>Additional Bartacks</u> The bartacks shall be 3/8 to 1/2 inch in length and positioned as follows: 1. One adjacent to the bottom stop of the front slide fastener, catching the slide fastener tapes and the protective fly. 2. One across the center front slide fastener tapes, centered between the bottom stop and the inseam, joining the edges of both tapes and catching the protective fly. 3. One horizontal bartack across the bottom of stitching on each side of the leg slide fastener covers close to the beading at the hemline. 4. One at the top the leg slide fastener covers adjacent to the slide fastener stop joining the beaded edges. 5. Two on each breast pocket, pencil pocket, thigh pocket and lower leg pocket; one at each end of the slide fastener cover adjacent to the slide fastener stop, joining the edges	bartack		28 per bartack

No.	Description of Operation	Stitch type	Seam & Stitching type	Stitches per inch
	of both slide fastener covers. The outer right thigh pocket bartack shall be midway between the stop and the outseam. 6. Two on each end of the back belt casing on the inside row of stitching. 7. One on each shoulder reinforcement piece at the top of the bi-swing back pleats on the outside row of stitching. 8. Two on the knife pocket at the top end, one on each side of the pocket opening on the binding. 9. One across the double row of stitching on the leg inseams and the sleeve side seams, 1/8 inch from the end of the sleeve and leg hems.			
35.	<u>Cleaning</u> Trim all thread ends on the outside of the coveralls to 1/8 - 1/4 inch or less and on the inside to 1/4 inch or less. Remove all loose thread ends on the outside and inside of coveralls. Remove all spots, stains, and shade tickets without injury to the fabric.			

3.7.2 Finished measurements. The finished coveralls shall conform to the measurements shown in Table VIII.

Table VIII. Finished Measurements in Inches

Coverall Sizes	1/2 Chest circ. (A)	1/2 Hip circ. (B)	Sleeve Inseam (C)	Center Front length (D)	Leg Inseam (E)	1/2 Thigh circ. (F)	1/2 Ankle circ. (G)
32S	18-3/8	19	21	26-1/4	28	12-1/8	8-7/8
34S	19-3/8	20-1/8	21-1/8	26-3/4	28-1/8	12-7/8	9-1/8
36S	20-3/8	21-1/4	21-1/4	27-3/8	28-1/4	13-5/8	9-1/4
38S	21-3/8	22-1/4	21-3/8	27-7/8	28-3/8	14-3/8	9-1/2
40S	22-3/8	23-1/4	21-1/2	28-3/8	28-1/2	15-1/8	9-5/8
42S	23-3/8	24-3/8	21-5/8	29	28-5/8	15-7/8	9-7/8
44S	24-3/8	25-3/8	21-3/4	29-5/8	28-3/4	16-5/8	10
46S	25-3/8	26-3/8	21-7/8	30-1/4	28-7/8	17-3/8	10-1/4
48S	26-3/8	27-1/2	22	30-7/8	29	18-1/8	10-3/8
50S	27-3/8	28-1/2	22-1/8	31-1/2	29-1/8	18-7/8	10-5/8
52S	28-3/8	29-5/8	22-1/4	32-1/8	29-1/4	19-5/8	10-7/8
32R	18-3/8	19	22-1/2	27-5/8	30	12-1/8	8-7/8
34R	19-3/8	20-1/8	22-5/8	28-1/4	30-1/8	12-7/8	9-1/8
36R	20-3/8	21-1/4	22-3/4	28-3/4	30-1/4	13-5/8	9-1/4
38R	21-3/8	22-1/4	22-7/8	29-3/8	30-3/8	14-3/8	9-1/2
40R	22-3/8	23-1/4	23	29-7/8	30-1/2	15-1/8	9-5/8
42R	23-3/8	24-3/8	23-1/8	30-3/8	30-5/8	15-7/8	9-7/8
44R	24-3/8	25-3/8	23 1/4	31	30-3/4	16-5/8	10
46R	25-3/8	26-3/8	23-3/8	31-5/8	30-7/8	17-3/8	10-1/4
48R	26-3/8	27-1/2	23-1/2	32-3/8	31	18-1/8	10-3/8
50R	27-3/8	28-1/2	23-5/8	32-7/8	31-1/8	18-7/8	10-5/8
52R	28-3/8	29-5/8	23-3/4	33-3/8	31-1/4	19-5/8	10-7/8
32L	18-3/8	19	24	29	32	12-1/8	8-7/8
34L	19-3/8	20-1/8	24-1/8	29-1/2	32-1/8	12-7/8	9-1/8
36L	20-3/8	21-1/4	24-1/4	30-1/8	32-1/4	13-5/8	9-1/4
38L	21-3/8	22-1/4	24-3/8	30-3/4	32-3/8	14-3/8	9-1/2
40L	22-3/8	23-1/4	24-1/2	31-3/8	32-1/2	15-1/8	9-5/8
42L	23-3/8	24-3/8	24-5/8	31-7/8	32-5/8	15-7/8	9-7/8
44L	24-3/8	25-3/8	24-3/4	33-3/8	32-3/4	16-5/8	10
46L	25-3/8	26-3/8	24-7/8	33-1/8	32-7/8	17-3/8	10-1/4
48L	26-3/8	27-1/2	25	33-3/4	33	18-1/8	10-3/8
50L	27-3/8	28-1/2	25-1/8	34-3/8	33-1/8	18-7/8	10-5/8
52L	28-3/8	29-5/8	25-1/4	34-7/8	33-1/4	19-5/8	10-7/8

Coverall Sizes	1/2 Chest circ. (A)	1/2 Hip circ. (B)	Sleeve Inseam (C)	Center Front length (D)	Leg Inseam (E)	1/2 Thigh circ. (F)	1/2 Ankle circ. (G)
32XL	18-3/8	19	25-1/2	30-3/8	34	12-1/8	8-7/8
34XL	19-3/8	20-1/8	25-5/8	30-7/8	34-1/8	12-7/8	9-1/8
36XL	20-3/8	21-1/4	25-3/4	31-1/2	34-1/4	13-5/8	9-1/4
38XL	21-3/8	22-1/4	25-7/8	32-1/8	34-3/8	14-3/8	9-1/2
40XL	22-3/8	23-1/4	26	32-3/4	34-1/2	15-1/8	9-5/8
42XL	23-3/8	24-3/8	26-1/8	33-1/4	34-5/8	15-7/8	9-7/8
44XL	24-3/8	25-3/8	26 1/4	33-3/4	34-3/4	16-5/8	10
46XL	25-3/8	26-3/8	26-3/8	34-3/8	34-7/8	17-3/8	10-1/4
48XL	26-3/8	27-1/2	26-1/2	35	35	18-1/8	10-3/8
50XL	27-3/8	28-1/2	26-5/8	35-5/8	35-1/8	18-7/8	10-5/8
52XL	28-3/8	29-5/8	26-3/4	36-1/4	35-1/4	19-5/8	10-7/8
32XXL	18-3/8	19	27	31-3/4	36	12-1/8	8-7/8
34XXL	19-3/8	20-1/8	27-1/8	32-1/4	36-1/8	12-7/8	9-1/8
36XXL	20-3/8	21-1/4	27-1/4	32-7/8	36-1/4	13-5/8	9-1/4
38XXL	21-3/8	22-1/4	27-3/8	33-1/2	36-3/8	14-3/8	9-1/2
40XXL	22-3/8	23-1/4	27-1/2	34-1/8	36-1/2	15-1/8	9-5/8
42XXL	23-3/8	24-3/8	27-5/8	34-5/8	36-5/8	15-7/8	9-7/8
44XXL	24-3/8	25-3/8	27-3/4	35-1/8	36-3/4	16-5/8	10
46XXL	25-3/8	26-3/8	27-7/8	35-5/8	36-7/8	17-3/8	10-1/4
48XXL	26-3/8	27-1/2	28	36-1/4	37	18-1/8	10-3/8
50XXL	27-3/8	28-1/2	28-1/8	36-7/8	37-1/8	18-7/8	10-5/8
52XXL	28-3/8	29-5/8	28-1/4	37-1/2	37-1/4	19-5/8	10-7/8
Tolerance	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$

Note: Sleeve pockets shall measure 4-1/2 inches in width (taken across the top of the pocket including the slide fastener cover) and 5-3/4 inches in length (taken along the side) for all sizes.

3.8 Guideline for taking finished measurements. All circumference measurements are taken from folded edge to folded edge with the garment laying flat. (This may not be seam to seam.)

- A. The **chest** is taken at the level of the junction of the underarm seam.
- B. The **hip measurement** is taken on the center seam of the lower back of the garment at a level 7-1/2 inches for a short, 8-1/2 inches for a regular and 9-1/2 inches for a long; down from the bottom of the back belt casing.
- C. **Sleeve inseam** taken from the junction point of the underarm sleeve seam and the side seam straight down the seam to the hem edge. (Both sleeve inseams shall be measured.)
- D. **Front length** is taken on the straight line of the center front from the neck to the crotch point. (To the top of the double lapped inseam.)
- E. The **leg inseam** is taken from the center of the beaded edges to the bottom of the hem. (Both leg inseams shall be measured.)
- F. The **thigh measurement** is taken in a horizontal line at the fullest part of the thigh above the top edge of the right thigh pocket from folded edge to folded edge. Note: This will generally be between 1-1/4 to 1-3/4 above the pocket and taken from outside folded edge to the center back crotch seam.
- G. The **ankle** is taken at bottom of hem with the slide fastener fully opened.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

- 1) First Article Inspection (see 4.2)
- 2) Quality conformance Inspection (see 4.3)

4.2 First Article inspection. The first article, submitted in accordance with 3.1, shall be inspected for compliance with design, configuration, workmanship and dimensional requirements. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4, except where otherwise indicated.

4.3.1 Knife pocket reinforcement webbing and lanyard cord. The webbing shall meet the requirements in paragraph 3.3.7 when tested for breaking strength in accordance with ASTM-D-5035. The cord shall meet the requirements in paragraph 3.3.6 when tested for breaking strength in accordance with ASTM-D-5035.

4.3.2 Hook and loop fastener tape test. The hook and loop fastener tape shall be tested in accordance with table IX:

TABLE IX. Hook and Loop Fastener Tape

Property/Characteristic	Test method	Requirement reference
Weight	ASTM-D-3776	Paragraph 3.3.3
Peel strength <u>1/</u> <u>3/</u>	AATCC 61, test 3A (after 3 launderings) ASTM-D-5170	Paragraph 3.3.3
Shear strength <u>1/</u> <u>2/</u> <u>3/</u>	AATCC 61, test 3A (after 3 launderings) ASTM-D-5169	Paragraph 3.3.3

1/ Samples containing splices shall not be used for the shear or peel test.

2/ For a 2 inch overlap.

3/ Tapes over 1 inch are slit to 1 inch prior to testing. Tapes of 1 inch or under are tested as is. Values reflect pounds per inch width for peel only. Shear values apply to a 2-inch overlap only.

4.3.3 Component and material inspection. Components and materials shall be tested in accordance with the requirements in this document.

4.3.4 Examination of end item. Examination of the end item shall be in accordance with the defects cited below:

TABLE X. Defects

	Major	Minor
<i>Material defects</i>		
Outside-Any hole, cut, tear, mend, burn, spot or stain, smash, misweave, Missing yarn, or weakening defect such as smash, multiple float needle chews, lo slub, shade part/bar, fabric cut on wrong side, stamping marks, color not as specified.	101	
Inside- Defects listed above not affecting the performance serviceability and appearance.		202
- Cloth stiffened, hardened or seared by heat (except where permitted)	102	

Cutting and marking		
-Any component part not cut in accordance with the directional lines on patterns	103	
-Any component part not cut in accordance with patterns that effects the design or finished dimensions	104	
-Corrosive metal fastening device or sewn on ticket or adhesive	105	
Seams and stitching		
-End of stitching not caught in other seams or not securely backstitched	106	202
-2 Broken stitches		
-More than 2 broken stitches	107	203
-2 skipped stitches		
-More than 2 skipped stitches	108	
-Loose tension resulting in loose seam	109	
-Thread breaks not securely overstitched for a 1/2 inch distance	110	204
-Seams which are twisted, or pleated		205
- Puckering-Double lapped seams with a rating of 2 in accordance with AATCC- 88B effecting a major portion of a seam.		
-Puckering-Double lapped seams with a rating of 1 in accordance with AATCC- 88B.	111	
- Puckering-Single needle seams with a rating of 2 in accordance with AATCC- 88B effecting a major portion of a seam.		206
- Puckering-Single needle seams with a rating of 1 in accordance with AATCC- 88B.	112	
-Edges not properly forced out		207
-Any joining seams staggered by 1/2 inch or more (underarms and leg inseams)	113	
-Open seam 1/2 inch or more		
-Open seam less than 1/2 inch	114	208
-Any raw edges 1/4 inch or more on outside of coverall		
-Any raw edges 1/4 inch or more on inside of coverall	115	209
-Any loose, missing or misplaced bartacks		
-Looper thread on outside of garment	116	210
-Edges of slide fastener covers overlapping or gapping more than 1/16 inch		211
-Thread ends not trimmed or removed		212
-Gage of stitching or stitches per inch irregular		213
Components		
-Any component part omitted, twisted, not specified size, missing, misplaced, poorly shaped, reversed or not cut in accordance with patterns effecting dimensions, design or finished dimensions.	117	
Metal components		
-Not specified type, size, or style; part of assembly omitted, bent or broken	118	
-Snap fastener distorted, cracked, sharp edges, nicks, burrs or metal slivers	119	
-Interference with movement of slide fastener	120	
-Any slide fastener not entirely covered by respective flap	121	
-Slide fastener buckling, causing rippling effect		214

Fronts -Uneven in length when slide fastener is closed by 1/2 inch or more when slide fastener is closed -Uneven in length when slide fastener is closed (up to 1/2 inch) -Thong not between 3 and 3-1/2 inches long	122	215 216
Collar -Collar not flush with center front edge (more than 1/4 inch) -Collar edges not forced out -Collar staggered more than 1/8 inch at neck edge -Points of diagonal stitching on undercollar not 2-1/2 to 3-1/2 inches apart	123 124	217 218
Pockets -Any pocket 1/2 inch or more out of horizontal or vertical alignment -Any pocket 3/8 to 1/2 inch out of horizontal or vertical alignment	125	219
Hook and Loop - Hook and loop out of alignment by 1/2 inch or more - Hook and loop out of alignment by up to 1/2 inch	126	220
Construction -Any operation not performed as specified.		221

5. PACKAGING

5.1 PACKAGING. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materials is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points packaging activity within the Military Departments Systems Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but it is not mandatory.)

6.1 Intended use. The coverall covered by this purchase description is intended to be worn as an outer garment by aircrew personnel.

6.2 Acquisition requirements. Acquisition documents must specify the following:

- a. Title, number and date of this document
- b. Types, classes and sizes required (see 1.2)
- c. Issue of DoDISS to be cited in the solicitation

- d. When first article is required (see 3.1, 4.2 and 6.2)
- e. Packaging Requirements (see 5.1)
- f. AQL and sampling provisions

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article shall be a preproduction sample. The contracting officer shall specify the appropriate type of first article and the number of units to be furnished. The contracting officer shall include specific instruction in all acquisition document regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing

Flight
Garment

MILITARY INTERESTS:

Custodians:

Air Force - 99 (ESA Type I)
Navy – AS (ESA Type II)
Army - GL

GSA-FSS

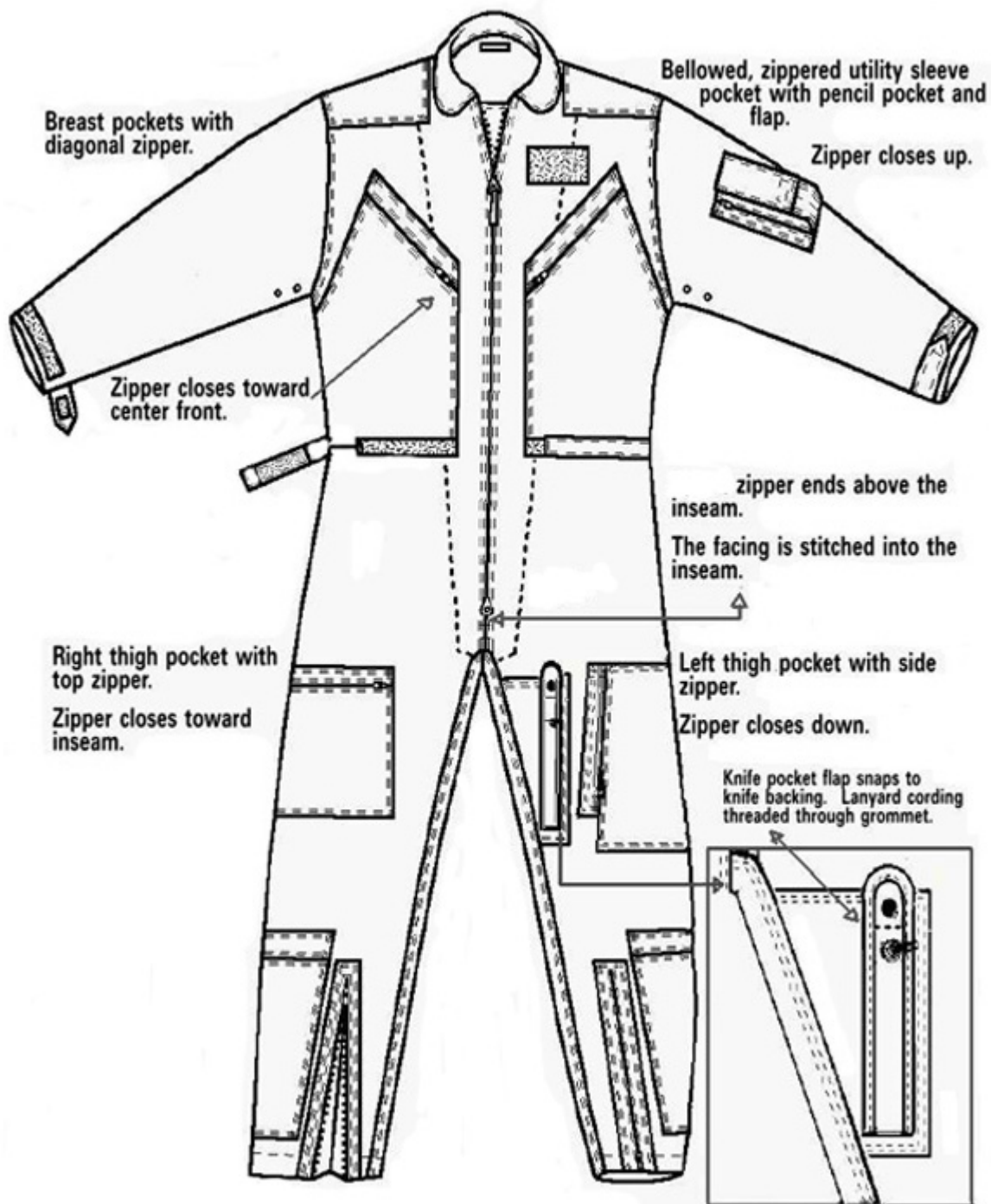
Review Activities:

Air Force - 11, 45
Navy - NU, MC

Preparing Activity:

DLA - CT

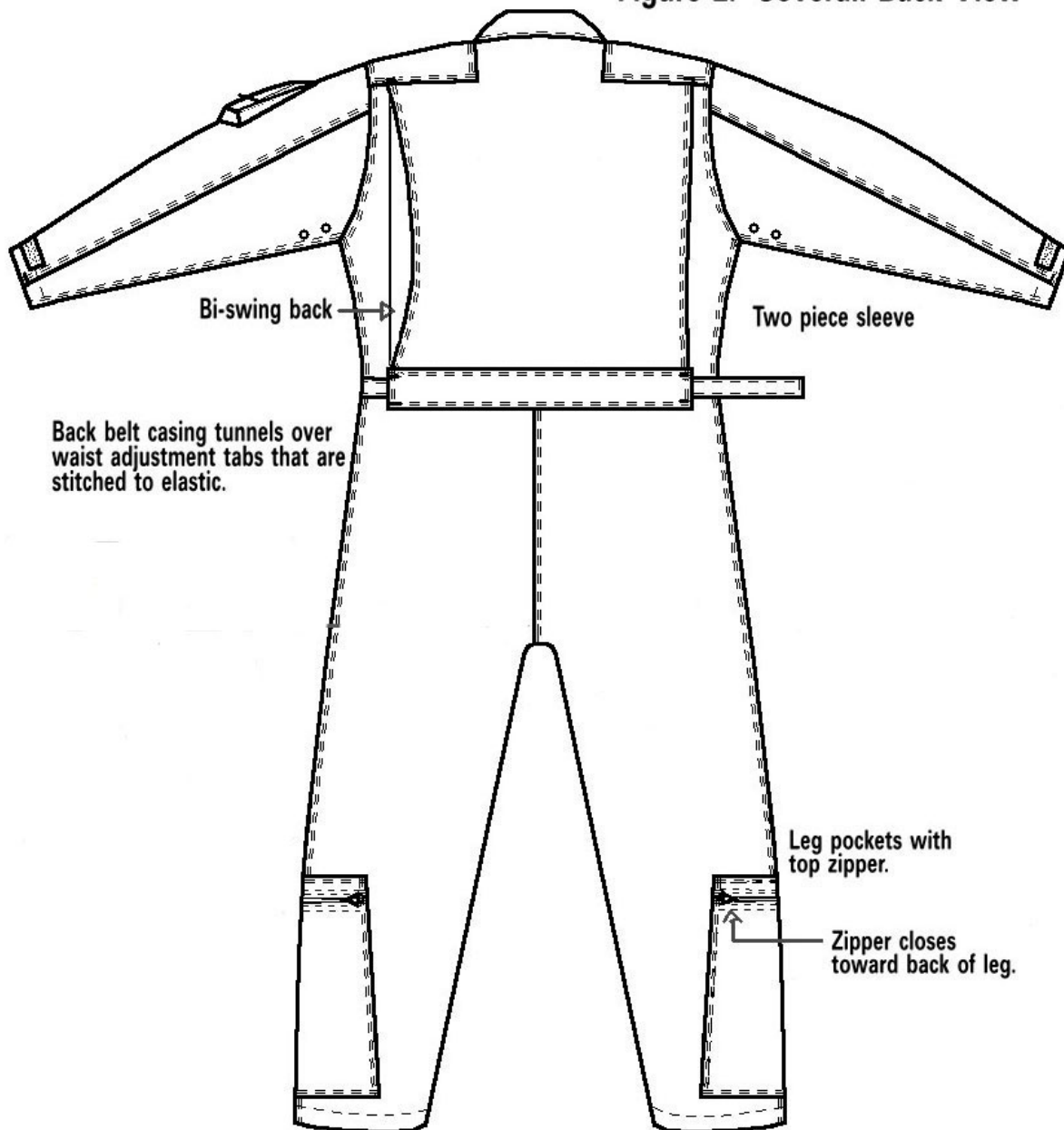
Figure 1. Coverall Front View



Knife pocket is stitched to the knife backing and assembly is then stitched to suit.

AFCO-WPAFB
svd: fs902

Figure 2. Coverall Back View



Bottom edges of pockets are 3-3/4" above finished hems at side seams and approximately 4" above finished hems at front and back corners.

Figure 3. Measurement and bartack placement guide

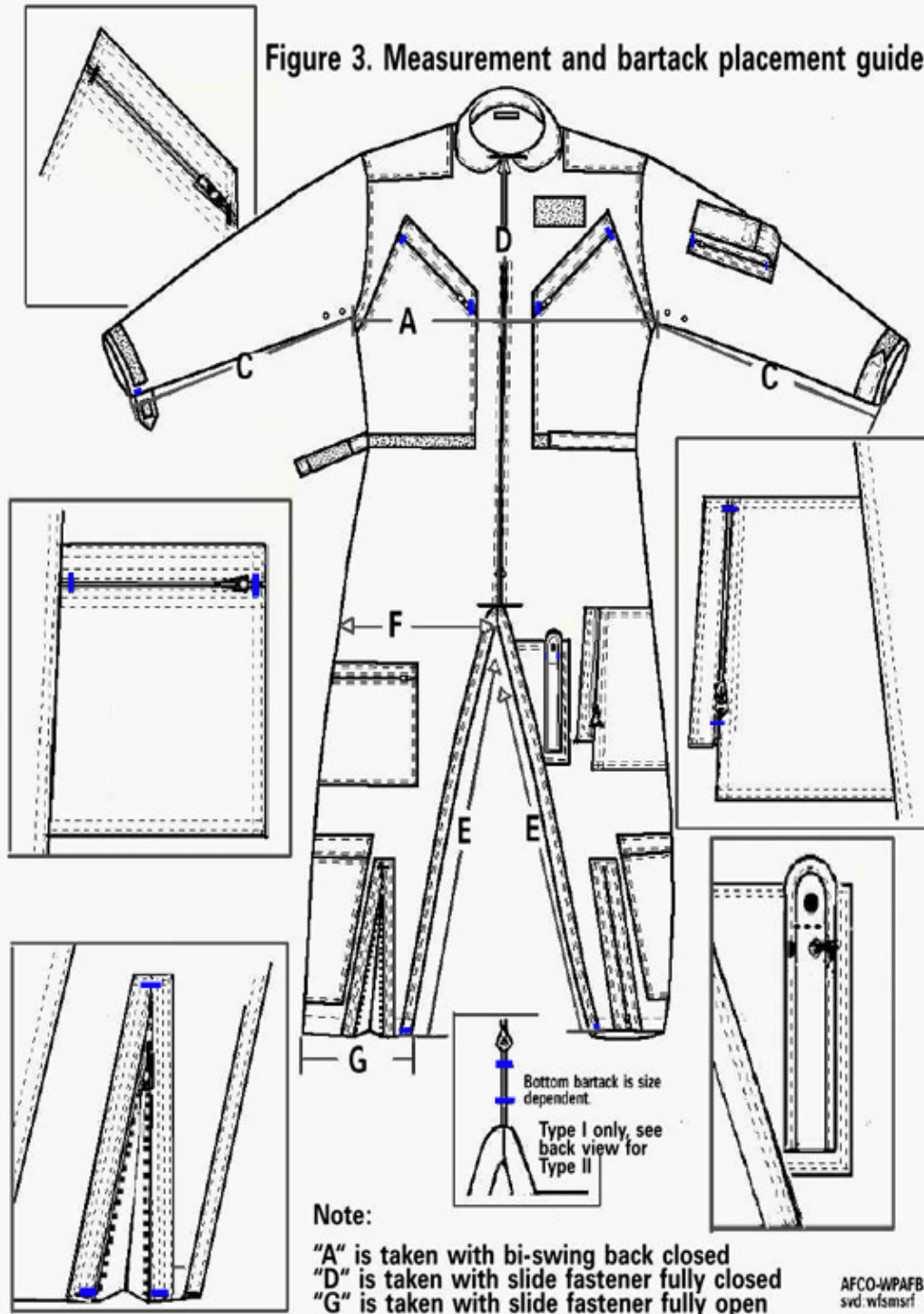


Figure 4. Measurement and bartack placement guide

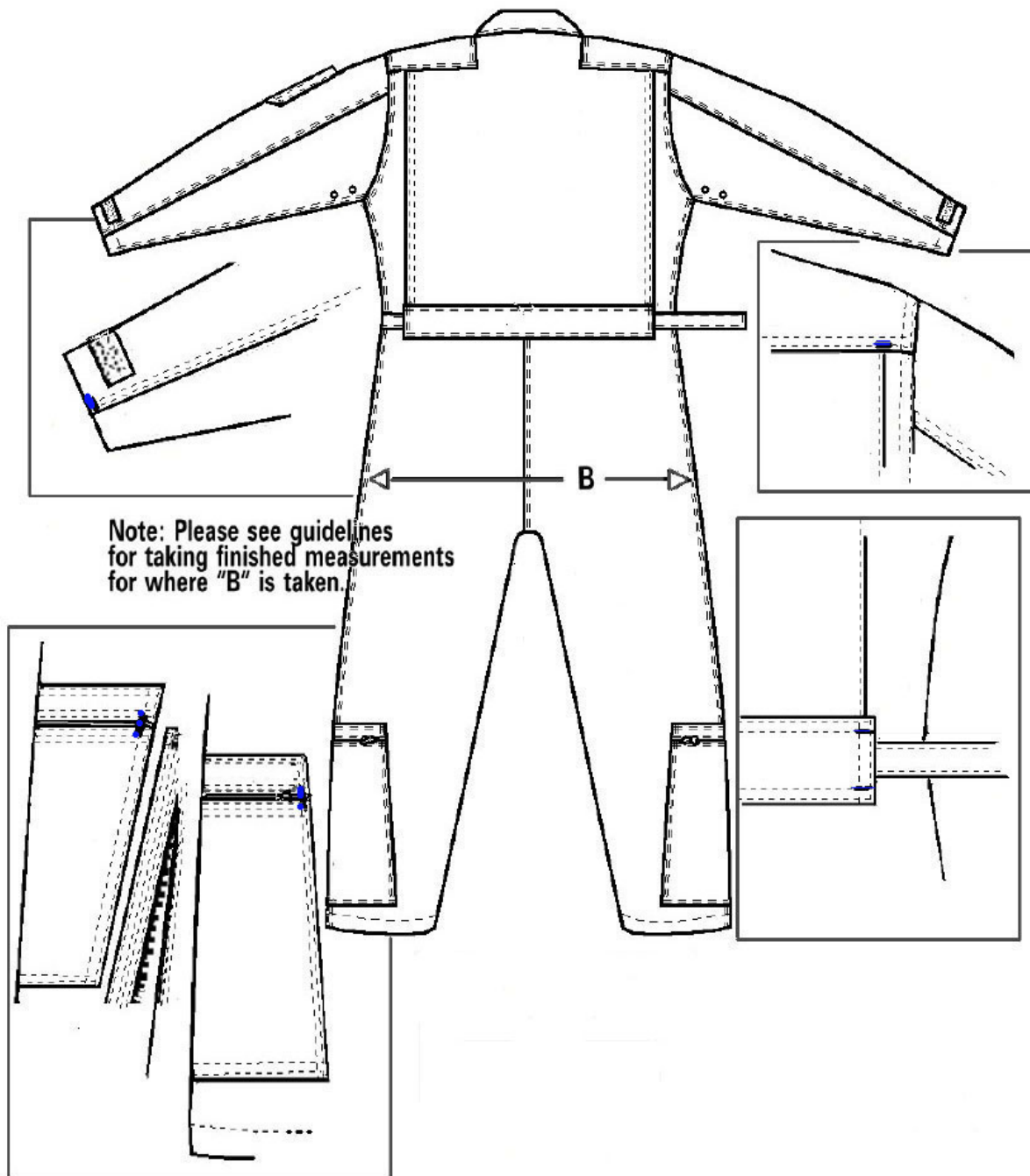


Figure 5., Sleeve Pocket/Pencil Pocket Configuration

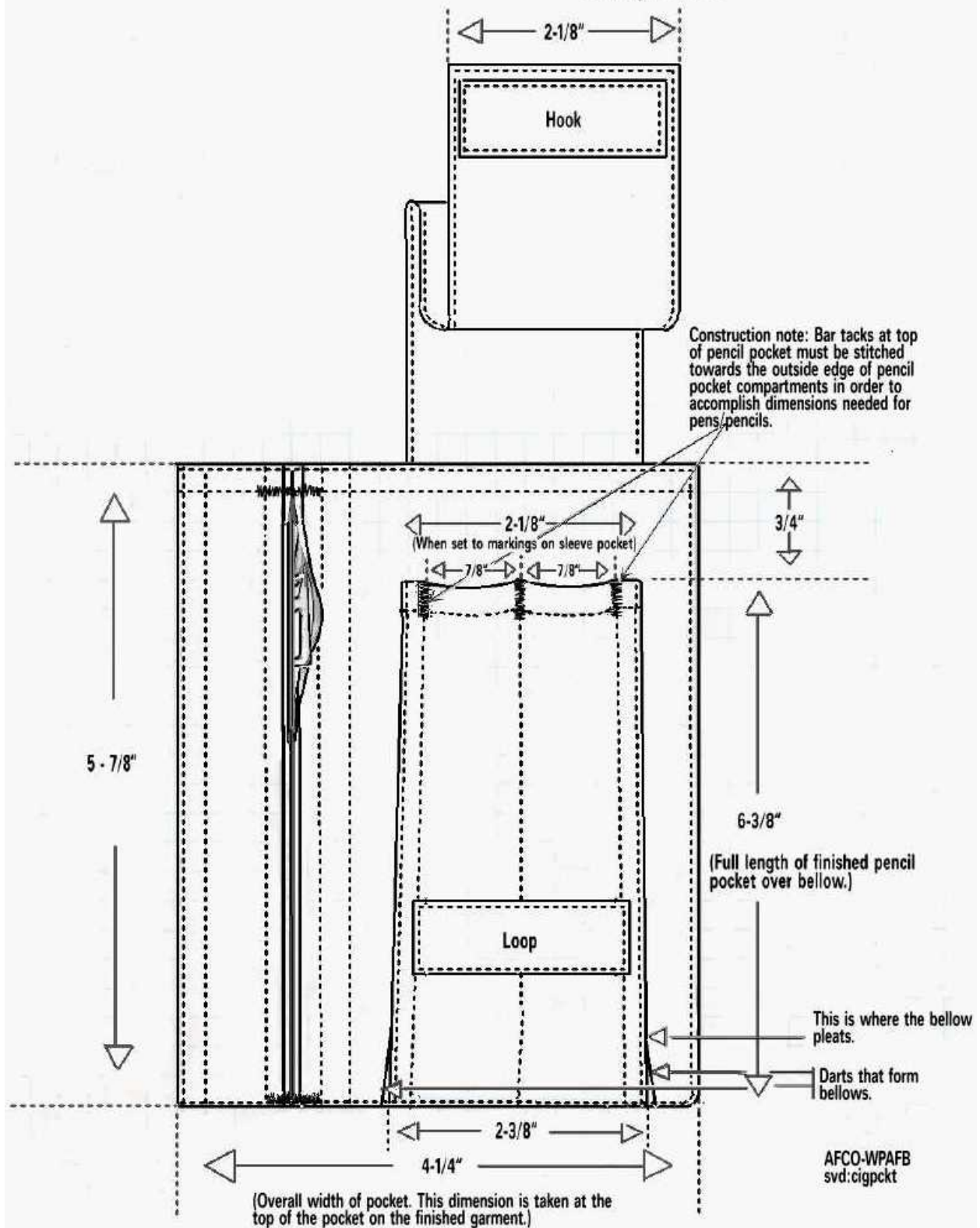
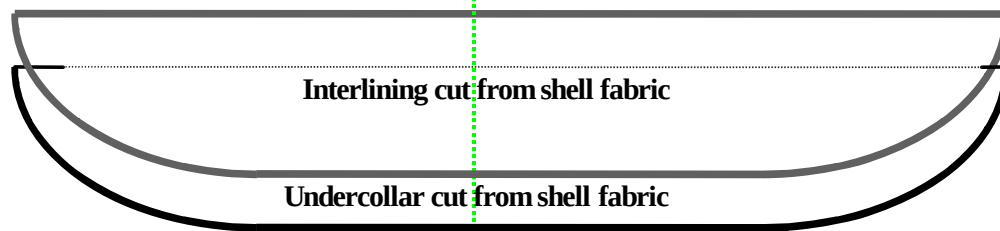
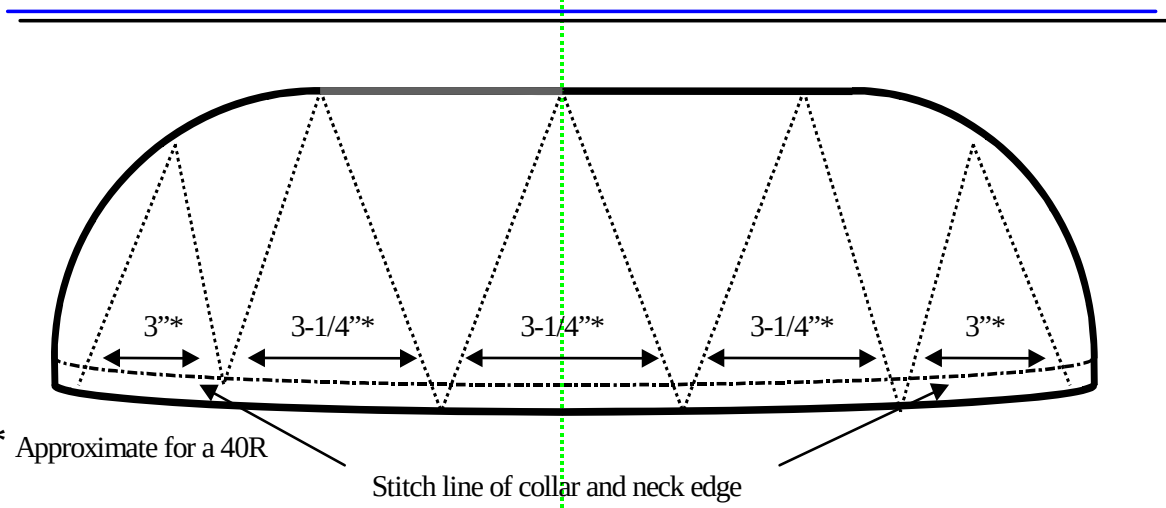


Figure 6. Collar

Collar, interlining and undercollar are all cut from shell fabric.

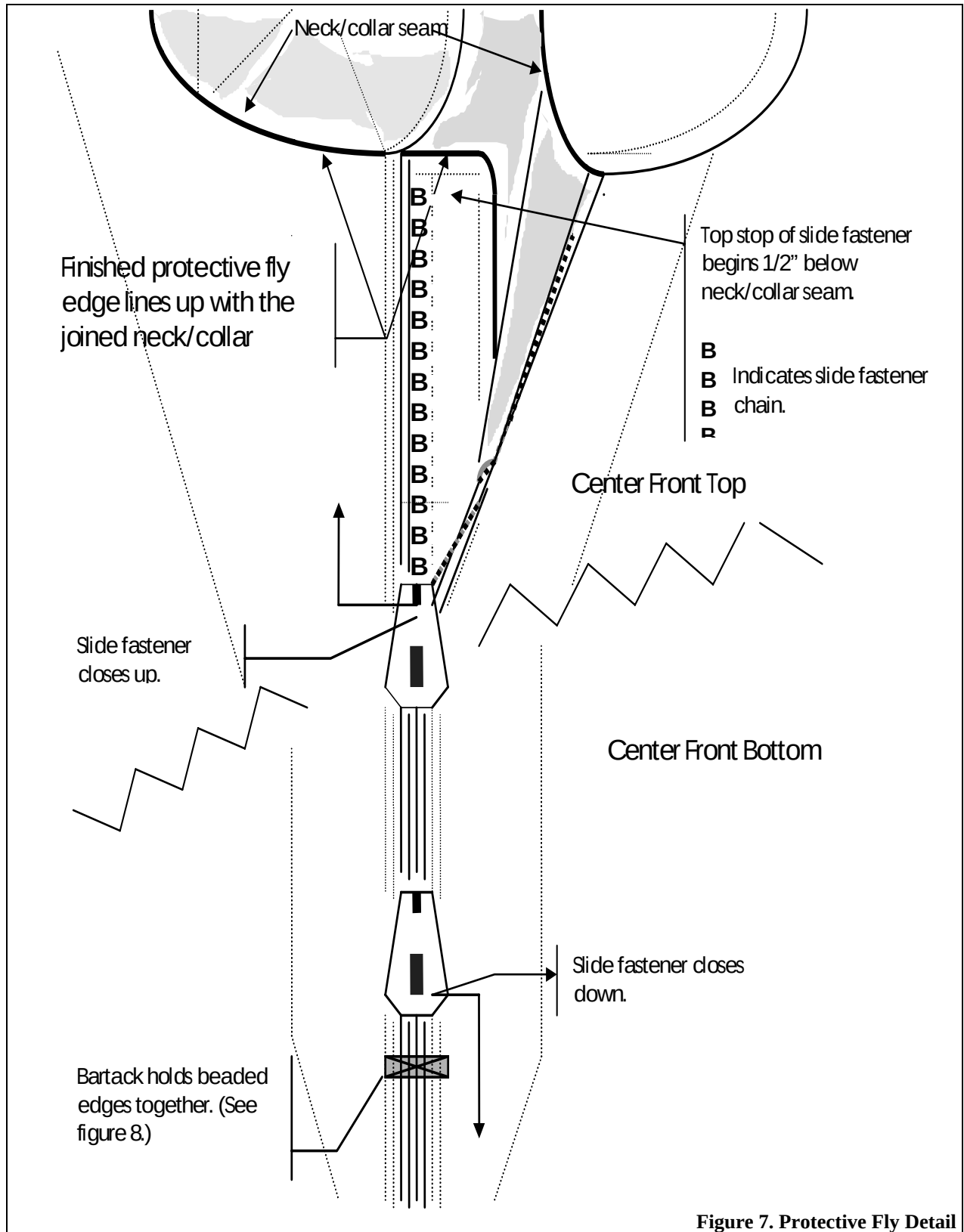


Stitch interlining to undercollar in the pattern shown below.



Finished view of undercollar quilted

The collar is then assembled to the top collar and is ready for attachment to neck edge.



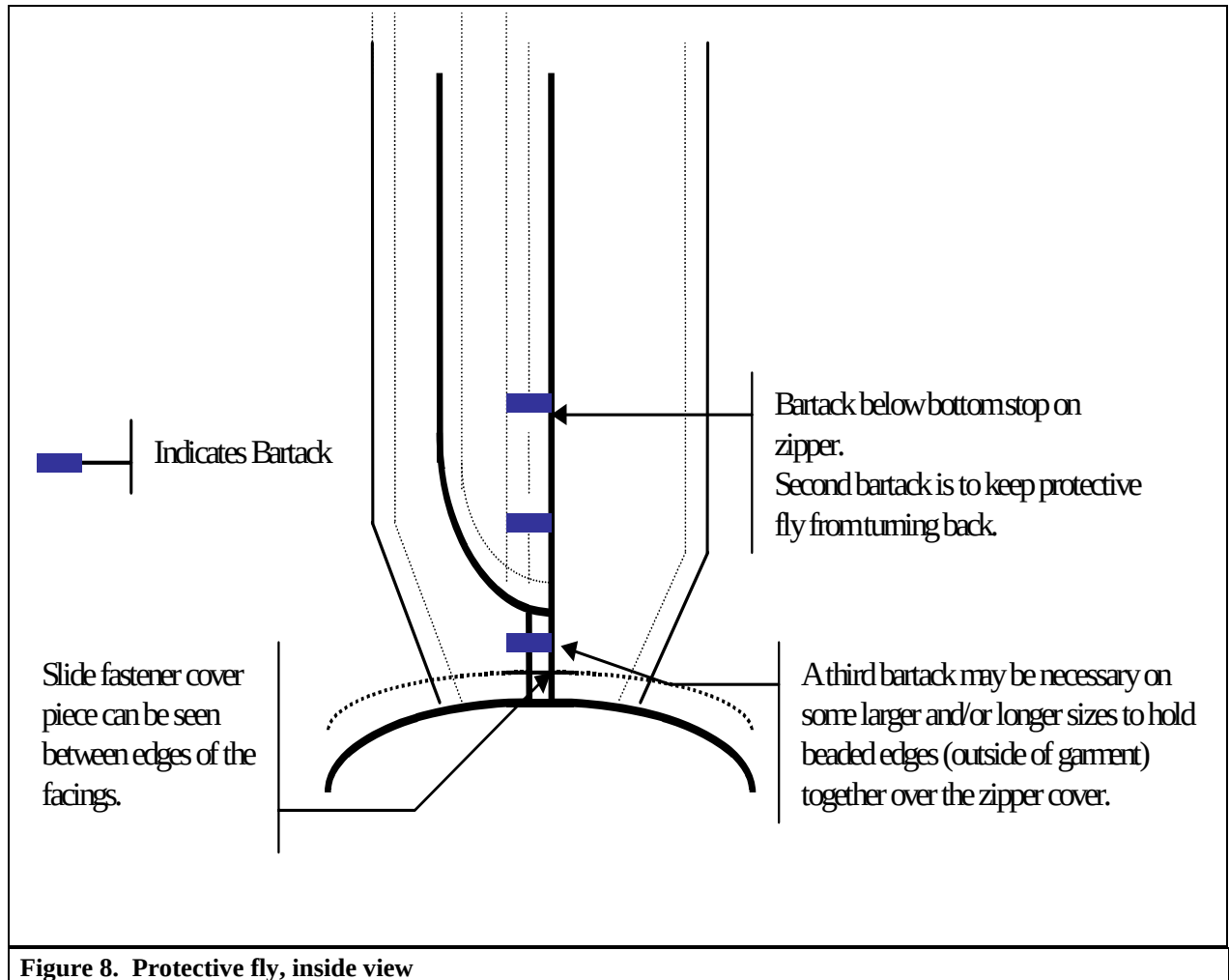


Figure 8. Protective fly, inside view

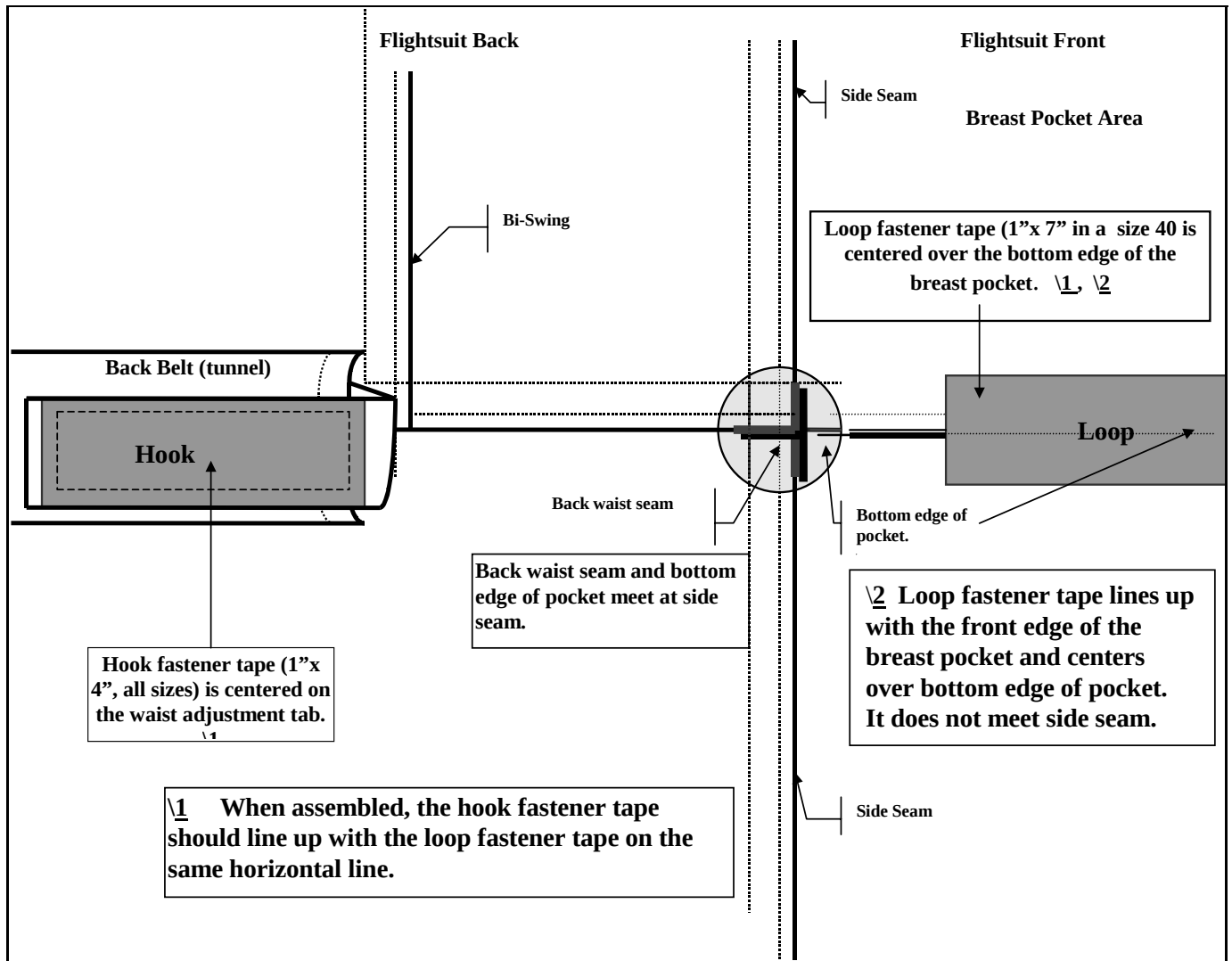
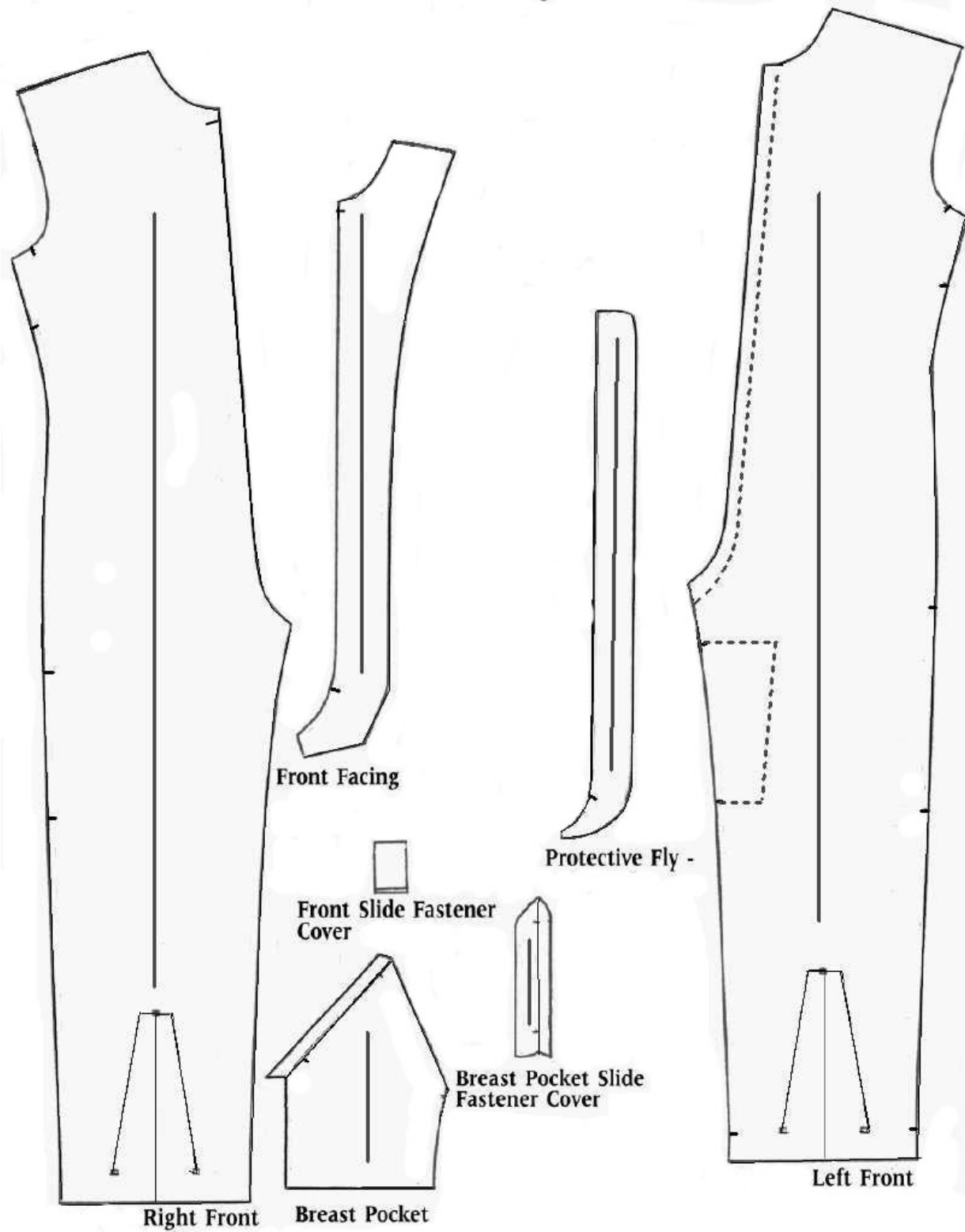


Figure 10. Detail: bottom of breast pocket to align with back waist seam

Figure 11. General Pattern Guide



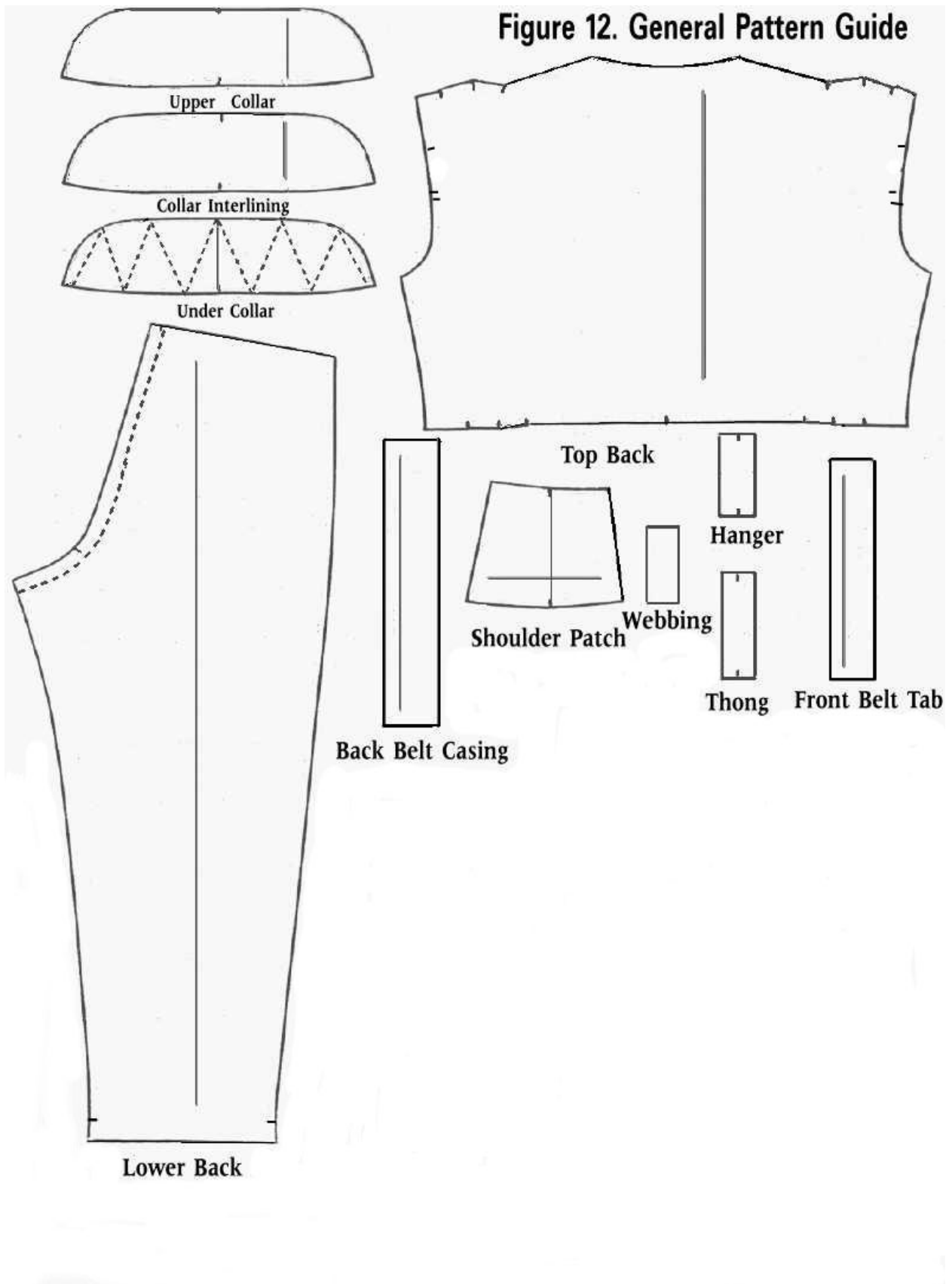


Figure 13. General Pattern Guide

